



SIMATS ENGINEERING



Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -2

CO2_AT2

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Course name: Computer Networks

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Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been

assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.

2. Identify the valid host IP address range for each subnet.

3. Calculate the number of usable host addresses available in each subnet.

4. Determine the remaining unused IP address range within the given network

block.

ANSWER :

Smart Campus Subnet Design

Base block allocated: 192.168.100.0/24

Subnet allocations using Variable Length Subnet Masking (VLSM) ordered from largest to smallest subnet size:

- School of Computing (/25)
 - Network Address: 192.168.100.0
 - Broadcast Address: 192.168.100.127
 - Valid Host IP Range: 192.168.100.1 to 192.168.100.126

- Usable Host Addresses: $2^{(32-25)} - 2 = 128 - 2 = 126$ addresses
- School of Electronics (/26)
 - Network Address: 192.168.100.128
 - Broadcast Address: 192.168.100.191
 - Valid Host IP Range: 192.168.100.129 to 192.168.100.190
 - Usable Host Addresses: $2^{(32-26)} - 2 = 64 - 2 = 62$ addresses
- School of Mechanical Engineering (/27)
 - Network Address: 192.168.100.192
 - Broadcast Address: 192.168.100.223
 - Valid Host IP Range: 192.168.100.193 to 192.168.100.222
 - Usable Host Addresses: $2^{(32-27)} - 2 = 32 - 2 = 30$ addresses
- Remaining Unused IP Address Range:
 - 192.168.100.224 to 192.168.100.255 (This represents an unused block of 32 addresses, or a /27 subnet block).

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid range of host IP addresses.**
- 3. Calculate the number of usable hosts in each subnet.**
- 4. Identify the unused IP address range remaining within the allocated IP block.**

ANSWER :

Headquarters Subnet Allocation

Base block allocated: 10.10.0.0/16

Sequential subnet allocation starting from 10.10.0.0:

- Software Development (/24)
 - Network Address: 10.10.0.0
 - Broadcast Address: 10.10.0.255
 - Valid Host IP Range: 10.10.0.1 to 10.10.0.254
 - Usable Host Addresses: $2^{(32-24)} - 2 = 256 - 2 = 254$ addresses
- Quality Assurance (/25)
 - Network Address: 10.10.1.0
 - Broadcast Address: 10.10.1.127
 - Valid Host IP Range: 10.10.1.1 to 10.10.1.126
 - Usable Host Addresses: $2^{(32-25)} - 2 = 128 - 2 = 126$ addresses
- Human Resources (/26)
 - Network Address: 10.10.1.128
 - Broadcast Address: 10.10.1.191
 - Valid Host IP Range: 10.10.1.129 to 10.10.1.190
 - Usable Host Addresses: $2^{(32-26)} - 2 = 64 - 2 = 62$ addresses
- Executive Management (/27)

- Network Address: 10.10.1.192
- Broadcast Address: 10.10.1.223
- Valid Host IP Range: 10.10.1.193 to 10.10.1.222
- Usable Host Addresses: $2^{(32-27)} - 2 = 32 - 2 = 30$ addresses
- Remaining Unused IP Address Range:
 - 10.10.1.224 to 10.10.255.255

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network: 2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**
- 3. Identify the network prefix using the /64 prefix length.**
- 4. Calculate the total number of host addresses supported within the subnet.**
- 5. the total number of possible host addresses available within this network.**

ANSWER:

Smart Hospital IPv6 Migration

Given IPv6 network block: 2001:0db8:abcd:0000:0000:0000:0000/64

1. Compressed Notation

Applying IPv6 compression rules (omitting leading zeros in each hextet and replacing the longest continuous sequence of zero hextets with ::):

- 2001:db8:abcd::/64

2. Full Hexadecimal Expansion

Restoring all 8 hextets and leading zeros:

- 2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix

Using the /64 prefix length:

- Network Prefix: 2001:db8:abcd::/64 (or in full form: 2001:0db8:abcd:0000::/64)

4. Total Usable Host Addresses

IPv6 host interface IDs use 64 bits ($128 - 64 = 64$). IPv6 does not use standard broadcast addresses, but the all-zero host address (::0) is reserved as the Subnet Router Anycast Address:

- Usable Hosts: $2^{64} - 1 = 18,446,744,073,709,551,615$ addresses

5. Total Possible Host Addresses

The total size of the 64-bit host space within this subnet:

- Total Possible Addresses: $2^{64} = 18,446,744,073,709,551,616$ addresses

4. An Internet Service Provider (ISP) connects multiple customer networks

through a core router. The router maintains the following routing table entries:

192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

A packet arrives at the router with the destination IP address: 192.168.2.45

As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.**
- 2. The matching routing table entry.**
- 3. The next-hop network/interface used to forward the packet.**
- 4. The default route that should be used if the destination does not match any routing table entry.**

ANSWER :

ISP Core Router Packet Forwarding

Destination IP address of incoming packet: 192.168.2.45

Routing Table Entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

1. Destination Subnet

Applying the /24 mask (255.255.255.0) to 192.168.2.45 gives:

- 192.168.2.0/24

2. Matching Routing Table Entry

Using the Longest Prefix Match rule, the incoming destination matches:

- 192.168.2.0/24

3. Next-Hop Network / Outgoing Interface

- The packet will be forwarded out of the outbound router interface associated with the 192.168.2.0/24 route (e.g., GigabitEthernet0/1 or the specific next-hop IP assigned to that subnet link).

4. Default Route

- If an incoming destination does not match any entry in the routing table, the router forwards the packet to the default route: 0.0.0.0/0 (Gateway of Last Resort). If no default route is configured, the packet is dropped and an ICMP Destination Unreachable message is returned.

5. A company establishes two office departments connected through a Layer-3 router.

The allocated subnet information is as follows: Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26 The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.**
- 2. Identify the valid host IP address range.**
- 3. Suggest suitable IP addresses for three computers in each LAN.**
- 4. Identify the default gateway that hosts in each LAN should use for communication.**

ANSWER:

Layer-3 Office Connectivity

Allocated subnets (/26 mask = 255.255.255.192, block size of 64 IPs):

- Administration LAN (192.168.10.0/26)
 - 1. Network & Broadcast Address:
 - Network Address: 192.168.10.0
 - Broadcast Address: 192.168.10.63
 - 2. Valid Host IP Address Range: 192.168.10.1 to 192.168.10.62
 - 3. Suitable IP Addresses for 3 Computers:
 - PC 1: 192.168.10.2
 - PC 2: 192.168.10.3
 - PC 3: 192.168.10.4
 - 4. Default Gateway: 192.168.10.1
- Finance LAN (192.168.10.64/26)
 - 1. Network & Broadcast Address:
 - Network Address: 192.168.10.64
 - Broadcast Address: 192.168.10.127
 - 2. Valid Host IP Address Range: 192.168.10.65 to 192.168.10.126
 - 3. Suitable IP Addresses for 3 Computers:
 - PC 1: 192.168.10.66
 - PC 2: 192.168.10.67
 - PC 3: 192.168.10.68
 - 4. Default Gateway: 192.168.10.65